

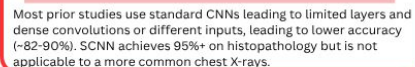
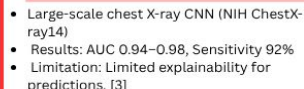
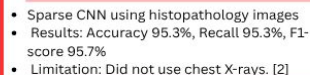


By: Aayush Katoch

- Lung cancer is often diagnosed too late because early tumors are difficult to detect on chest X-rays: ~20–30% of cases are missed during early stages. [5][12]
- Radiologists may struggle to distinguish early cancer from normal anatomy, leading to false negatives. [5]
- Existing CNN-based systems often function as “black boxes” without visual explanations. [2]
- Lack of explainability limits clinical trust and real-world adoption of AI systems. [2]

- Lung cancer is the leading cause of cancer-related deaths worldwide. [9][11]
- Early-stage tumors are difficult to identify due to low contrast and overlapping anatomy. [5]
- Chest X-rays are inexpensive and widely available compared to CT scans. [6]
- CNNs learn hierarchical image features for abnormality detection. [2]
- Clinical feature fusion combines imaging with patient information to improve predictions.

- CNN-based CAD system using JSRT dataset
- Results: Accuracy 83.2%, Sensitivity 83.7%, Specificity 82.8%
- Limitation: Small dataset caused overfitting [1]



SYSTEM ARCHITECTURE

224x224x3

DenseNet121 (pretrained) CNN architecture

Dense Block (1)
Conv
pooling

Transition Layer (1)

Dense Block (2)
Conv
pooling

Transition Layer (2)

Dense Block (3)
Conv
pooling

Transition Layer (3)

Dense Block (4)
Conv
pooling

Grad-Coll Heatmap
activations

Concatenate

Dense (4)

Sigmoid Output

Cancer probability

<Gender, age>

Dense (16)

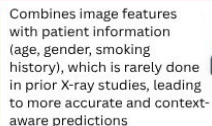
Concatenate

MLP Branch

*Diagram continued on the next

Dataset	Preprocessing	Training Details
- NIH Chest X-ray dataset used as primary source. [14] 5,606 total chest X-ray images ~54% "No Finding" images and ~46% disease-positive cases - Dataset split into training (70%), validation (15%), and testing (15%)	- Resized all images to 224x224 pixels - Normalized pixel values to range [0,1] - Binary labels: - Normal ("No Finding") = 0 - Disease ("Cancer") = 1	- Backbone: DenseNet121 pretrained on ImageNet [2] - Optimizer: Adam - Loss: Binary Cross Entropy - Batch size: 32 - Epochs: 27 - Early stopping based on validation loss

Uses multi-level attention and Grad-CAM to highlight regions that influenced the model's decision, making predictions transparent and easier for doctors to interpret



- Provides explainable AI support for radiologists by highlighting suspicious chest regions.



Helps reduce missed diagnoses and improve early lung cancer detection.



- Evaluate using Accuracy, Sensitivity, and AUC
- Reader studies with vs. without AI assistance

The diagram illustrates the proposed Grad-CAM++ architecture. It starts with an input image being processed by a CNN to generate feature maps. These feature maps are then used to calculate weights w_i^c for each class c . The weights are summed and passed through a ReLU activation to produce the final Grad-CAM++ heatmap. This heatmap is then multiplied element-wise with the original feature maps to produce the guided class activation map. The diagram also shows a 'Guided backpropagation' path from the final output back to the input image.

Confusion Matrix

	Normal	Cancer
Normal	820	35
Cancer	22	123

Key Metrics

- Accuracy: **93.0%**
- Sensitivity (Recall): **91.6%**
- Specificity: **93.2%**
- Precision: **91.7%**
- F1-score: **91.6%**
- AUROC: **0.97**

ROC Curve

The ROC curve shows the True Positive Rate (Y-axis) versus the False Positive Rate (X-axis). The curve is significantly above the diagonal line, indicating good model performance. The AUROC is 0.95.

- Early-stage nodules remain difficult to detect.
- Dataset bias may limit generalization to diverse populations.
- Clinical data limited to age and gender.
- Grad-CAM explanations are approximate visualizations.

- Add more clinical features (smoking history, symptoms, medical records)
- Use larger multi-center datasets
- Extend to multi-class disease classification
- Improve Grad-CAM localization quality